Evidence-Preserving Graph Denoising for Noisy-Label Intrusion Detection: A Leakage-Audited Study of When Graph Structure Helps

Abstract

Intrusion-detection classifiers are increasingly trained on labels that are delayed, rule-generated, or inconsistent across telemetry sources. A common remedy is dataset purification: detect suspicious labels and *delete* them before training. In security operations this is risky, because the deleted instances are frequently the rare, boundary attack samples that analysts most need. We argue that in security machine learning, suspicious does not mean useless: samples flagged as noisy may be operational evidence of rare or emerging threats, and we therefore recast label denoising as *evidence-preserving data curation* rather than purification. We revisit CoLD, a recent collaborative label-denoising framework, and ask whether lifting it to a multi-view graph and replacing hard deletion with *evidence-preserving soft weighting* improves noisy-label intrusion detection. Our study makes three contributions grounded in reproducible, real-data experiments on CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15. First, we show that a label-space graph consistency diagnostic (Graph-CDM) built over host, IP, temporal, and where-available behavioral views yields a modest but statistically significant aggregate Macro-F1 improvement over the CoLD baseline (Holm-corrected $p < 0.05$ on all three datasets; up to $+0.11$ Macro-F1 on CICIDS-2017 but $\leq +0.01$ on the other two datasets under the final protocol), with the gain concentrated where graph views carry discriminative structure. Second, within a shared pipeline, we find that evidence-preserving soft weighting is statistically indistinguishable from hard deletion on aggregate Macro-F1, yet significantly improves *rare/tail attack-class* Macro-F1 under high asymmetric label noise (ΔMacro-F1 = +0.073, +0.022, +0.040 on CICIDS/CESNET/UNSW; Holm $p < 10^{-4}$; Cohen's $d_z \in [0.91, 1.29]$), at no aggregate cost. Benchmarked under the identical protocol, an external Confident Learning anchor outperforms our weighting on aggregate and tail metrics on all three datasets, while full Co-Teaching collapses on tail classes—confirming empirically that deletion- and forgetting-based mechanisms can sacrifice rare-attack evidence unless the noise diagnostic is precise. Third, we contribute a leakage audit showing that naive graph denoising can be dramatically inflated by duplicate records (24% exact duplicates in CICIDS-2017) and by any use of clean-label signal; removing these collapses an apparent +28 percentage-point advantage to a defensible +6.7 points. We release the audit, the graph-informativeness diagnostic whose scores are consistent with when the approach helps, and the full reproducibility package. The result is a bounded, honest claim: evidence-preserving retention is a mechanism-level remedy for rare-attack evidence loss within a shared pipeline, not a universal accuracy improvement; confident-learning-precision pruning is the performance bar that graph-based denoising must still clear.

Keywords: intrusion detection, noisy labels, graph learning, evidence preservation, security operations, rare-class robustness

1 Introduction

Machine-learning classifiers underpin modern network intrusion-detection systems (IDS) and security operations centers (SOCs). These models are trained from labels that are seldom perfect: they may be delayed, produced by rules of uneven quality, or shaped by analyst workflows that emphasize high-severity cases [2, 3]. The resulting label noise degrades both detection quality and operational triage. Crucially, the operational cost of noise is asymmetric. Discarding a mislabeled benign flow may raise a benchmark score, but discarding an informative low-frequency attack sample removes exactly the evidence an analyst needs to investigate a rare or emerging threat.

This asymmetry motivates a reframing of the problem. Conventional noisy-label pipelines treat suspicious samples as contamination to be purged; we instead treat security data curation as an *evidence-preservation* problem, in which the decision to delete must weigh the probability that a sample is mislabeled against the operational cost of losing it. In a SOC, the samples most likely to be flagged as inconsistent—rare attacks whose feature neighborhoods overlap benign traffic—are precisely those whose deletion is most costly. The question is therefore not only *how accurately* noise can be detected, but *what should be done* with samples the detector cannot confidently classify: delete, trust, or retain at reduced influence.

CoLD [1] recently framed noisy-label IDS through *local consistency*: because different traffic classes can share feature neighborhoods, label noise induces spurious associations, and a collaborative representation can identify and remove likely-noisy samples before training. CoLD, like most dataset-purification methods, ultimately performs *hard deletion*. We take CoLD as our baseline and ask two questions. (Q1) Does lifting the diagnostic from independent samples to a multi-view graph, kept strictly in label space, improve robustness? (Q2) Does replacing hard deletion with an evidence-preserving soft weight—one that down-weights rather than deletes suspicious-but-informative samples—preserve rare-attack evidence and improve operational detection?

Answering these questions honestly required an unusual amount of methodological hygiene. Our first implementations produced spectacular gains that did not survive scrutiny: a leakage audit revealed that (i) CICIDS-2017 contains 24% exact-duplicate records that let neighborhood consistency trivially recover labels, and (ii) any use of clean-label signal in the evidence or graph construction inflates results. After removing both, the apparent advantage shrank to a modest, defensible level, and the headline metric we had used for evidence retention turned out to be tautological. We report this process because it is, we believe, a contribution in its own right: graph-based denoising results in security are easy to inflate and must be audited.

Contributions.

Together, the following contributions instantiate an evidence-preserving data-curation framing for security machine learning, in which suspicious samples are treated as potentially valuable evidence rather than disposable noise:

1. A *label-space* multi-view graph consistency diagnostic (Graph-CDM) that extends CoLD to host/IP/temporal (and, where available, behavioral) views and delivers a modest but Holm-significant aggregate Macro-F1 gain over CoLD on three real datasets (Section 6.1).
2. An *evidence-preserving soft weighting* rule that is indistinguishable from hard deletion on aggregate accuracy yet significantly improves rare/tail attack-class Macro-F1 under high asymmetric noise (Section 6.3)—a mechanism-level benefit within the shared pipeline. Benchmarked under the identical protocol, external anchors bound this benefit from both sides: confident-learning pruning sets a higher performance bar on every dataset, while full co-teaching's forgetting discards the rare classes outright.
3. A *leakage audit* and a *graph-informativeness diagnostic*: we quantify duplicate- and oracle-driven inflation, and show that a per-dataset neighborhood-homophily measure is consistent with when the graph helps (three datasets; descriptive, not inferential) (Sections 6.4–6.5).

We deliberately bound our claims. Graph-CoLD is not a universal accuracy improvement; its distinctive value is preserving rare-attack evidence that hard deletion discards.

2 Related Work

Noisy-label learning.

Approaches divide into robust-training and dataset-purification families. Robust training modifies losses or training dynamics: loss correction with a transition matrix [4], peer-network small-loss selection (Co-teaching [5], Co-teaching+ [6]), disagreement-based updates (Decoupling [7]), and semi-supervised treatment of suspected-noisy data (DivideMix [8]). Purification detects and removes mislabeled instances: confident learning [9], representation-based filtering (FINE [10]), and, in security, MORSE [11] and MCRe [12]. CoLD [1] is the closest prior work and our baseline. Our departure is twofold: we keep the diagnostic in label space (avoiding the distance-metric fragility CoLD itself criticizes), and we replace deletion with evidence-preserving weighting motivated by the semi-supervised insight that suspected-noisy samples carry usable signal [8, 11].

Graph learning for security telemetry.

Security events are relational, and graph neural networks [13–16] naturally propagate contextual evidence. Provenance-graph IDS such as Flash [18] and Argus [19] demonstrate the value of structure for enterprise detection. We use graph representation learning as a backbone but, unlike embedding-distance denoisers, keep the noise diagnostic in label space so it stays aligned with the classifier.

SOC alert handling and concept drift.

Operational systems must also cope with alert volume and distribution shift. Alert-classification and triage work [21, 22] and drift-aware detection [20] motivate our operational framing, though our contribution is denoising for rare-attack retention rather than a deployed triage study.

Contrastive self-supervision.

Our representation stage uses cross-view contrastive learning [17], adapted so that the same node in two views forms a positive pair.

2.1 Positioning and Open Challenges

Noisy-label learning for intrusion detection

Learning from noisy labels has become an important challenge in security analytics because network telemetry is often collected through heterogeneous sources, automated rules, and delayed analyst feedback. Unlike conventional computer vision benchmarks where label corruption can often be controlled synthetically, security labels may reflect incomplete attack knowledge, imbalanced detection rules, and evolving adversarial behaviors [26, 27]. Therefore, improving robustness requires not only higher classification accuracy but also understanding which samples should be trusted and which samples contain useful but uncertain evidence.

Existing noisy-label learning methods can be broadly divided into robust training and dataset purification. Robust training approaches attempt to reduce the influence of corrupted samples during optimization. Loss correction methods estimate label transition characteristics and modify the training objective [28]. Sample-selection approaches exploit uncertainty, loss dynamics, or disagreement between models to identify potentially noisy instances [34, 35]. Co-teaching and related disagreement-based approaches train multiple networks and exchange selected small-loss samples to reduce memorization of corrupted labels [5, 6]. Decoupling further separates the decision of when to update from how to update, demonstrating that disagreement between classifiers can provide useful information under label corruption [7].

A different family focuses on dataset purification. Confident Learning estimates label uncertainty by identifying statistically inconsistent annotations [9]. FINE explores representation-space filtering of noisy instances through latent feature structures [10]. In security domains, MCRe and MORSE extend noisy-label handling toward malicious traffic and malware classification scenarios [11, 12]. More recently, intrusion detection studies have explored contrastive learning and representation enhancement as a way to improve robustness against noisy or incomplete observations [29, 30].

CoLD provides the closest foundation for our work by introducing collaborative label denoising specifically for network intrusion detection. However, existing purification-oriented methods generally treat suspicious samples as removable noise. This assumption is problematic in SOC environments because rare attack samples may simultaneously appear inconsistent and operationally valuable. Graph-CoLD therefore does not aim to replace purification, but instead explores whether structural context and evidence-preserving weighting can retain informative suspicious instances.

Graph learning and contrastive representation for security

Graph representation learning has become increasingly important for security analytics because network activities naturally form relational structures among hosts, addresses, sessions, and temporal events. Graph convolutional networks and graph attention networks demonstrated that neighborhood information can improve representation learning when samples are connected through meaningful relations [13, 14]. Inductive graph representation learning further enables scalable modeling of previously unseen nodes [15], while heterogeneous graph transformers extend graph reasoning to multiple relation types [16].

In security applications, graph methods have mainly focused on attack detection, provenance analysis, and contextual behavior modeling. Systems such as Flash demonstrate the value of provenance graph representation learning for enterprise intrusion detection [18]. Other work investigates security-oriented representation learning and graph analytics for detecting suspicious behaviors [19]. These approaches usually focus on discovering malicious structures from graph embeddings.

Graph-CoLD differs from these methods in its objective. The graph structure is not directly used as an anomaly detector; instead, it provides contextual evidence for a label-space consistency diagnostic. This distinction is important because embedding distance alone may capture feature similarity while remaining disconnected from the classifier's final prediction behavior. By combining graph context with prediction-space disagreement, Graph-CoLD aims to identify unreliable labels while preserving samples that remain informative.

SOC alert triage, prioritization, and concept drift

Modern SOC environments face not only detection accuracy challenges but also large-scale alert management problems. Excessive alerts, evolving attack behaviors, and changing network distributions create operational pressure for analysts. Alert classification and active-learning approaches have explored ways to reduce analyst workload by improving alert categorization [21]. Reinforcement-learning-based alert prioritization methods further investigate how automated systems can rank alerts according to operational importance [22].

Beyond static classification, concept drift has become a critical challenge in long-running security monitoring. Detection systems must adapt when normal traffic patterns, attack strategies, and service behaviors evolve. Studies such as CADE, RecDA, and Insomnia investigate drift detection and adaptation mechanisms for network intrusion detection [20, 36, 37]. Recent SOC-oriented research also explores alert aggregation and filtering through deep learning models to improve investigation efficiency [31–33].

These operational studies motivate our security framing, but Graph-CoLD addresses a different layer of the problem. Rather than replacing analyst triage or predicting attack priority directly, we focus on improving the quality of the training evidence that supports downstream detection. The key difference is that Graph-CoLD combines three principles: label-space consistency diagnosis, evidence-preserving retention, and leakage-audited evaluation. This design aims to answer when graph-based denoising genuinely helps security analytics and when structural information is insufficient.

3 Problem Formulation

Let V be the set of training flows and $\mathcal{M}$ the set of *active* graph views. Each view $m \in \mathcal{M}$ induces a graph $G^m = (V, E^m)$ over the same nodes. The observed training label $\tilde{y}_v$ may differ from the unobserved clean label y_v. We seek a classifier that (i) resists the effect of noisy labels, and (ii) preserves operationally useful evidence—especially clean samples of rare attack classes that a purifier is tempted to discard.

We adopt three design principles. First, the noise diagnostic operates in *label/prediction space*, not embedding distance, so it remains aligned with the classifier and does not re-introduce the local-consistency fragility of distance-based purifiers. Second, denoising is *evidence preserving*: a suspicious but informative sample is down-weighted rather than removed. Third, the evaluation is traceable to audited real data, with explicit scope limits.

4 Method

Graph-CoLD is a two-stage extension of CoLD: self-supervised multi-view representation learning, followed by a label-space consistency diagnostic that drives an evidence-preserving weight (Figure 1).

4.1 Multi-view graph construction

Views are built from semantically related feature blocks over the training nodes: a *host* view links samples sharing endpoint identifiers; an *IP* view uses communication/flow

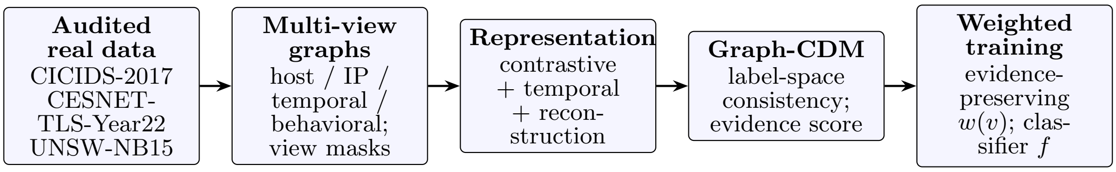

Fig. 1 Graph-CoLD pipeline. Only views supported by a dataset's verified schema are activated; unsupported views are disabled rather than fabricated. The consistency diagnostic (Graph-CDM) is computed in label space and feeds an evidence-preserving soft weight and a downstream classifier. Datasets are CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15 (active views vary by dataset; see Table 1).

statistics, ports, protocol and TLS metadata; a *temporal* view links samples within a time window; a *behavioral* view groups service, protocol, port-state, and related behavioral feature blocks (it is a feature-block view, not host-process lineage) and is used only where such fields exist. Support is part of the reproducible dataset contract: CICIDS-2017 activates host/IP/temporal, CESNET-TLS-Year22 activates IP/temporal, and UNSW-NB15 activates temporal/behavioral. Threat-intelligence views are disabled when the fields are absent. Crucially, edges never cross the train/test split, and (Section 6.4) exact/near-duplicate edges are removed.

4.2 Self-supervised representation

Each active view produces a node embedding z_v^m from Bernoulli-masked input features, $x'_v = x_v \odot \text{Bernoulli}(1-\delta)$. Cross-view contrastive learning treats the same node in two views as a positive pair and other batch nodes as negatives:

$$\mathcal{L}_{\text{con}} = -\log \frac{\exp(\text{sim}(z_v^a, z_v^b)/\tau)}{\sum_u \exp(\text{sim}(z_v, z_u)/\tau)}. \tag{1}$$

Temporal alignment and reconstruction terms regularize the embedding, $\mathcal{L}_{\text{rep}} = \mathcal{L}_{\text{con}} + \alpha\mathcal{L}_{\text{temporal}} + \beta\mathcal{L}_{\text{recon}}$; active views are fused by mean aggregation to obtain z_v.

4.3 Graph-CDM: a label-space consistency diagnostic

For each node v,

$$\mathrm{GraphCDM}(v) = \lambda_1 D_{\mathrm{pred}}(v) + \lambda_2 D_{\mathrm{neigh}}(v) + \lambda_3 D_{\mathrm{view}}(v) + \lambda_4 D_{\mathrm{chain}}(v), \tag{2}$$

where the prediction term compares per-view predicted labels with the observed *noisy* training label,

$$D_{\mathrm{pred}}(v) = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}[\tilde{y}_v^m \neq \tilde{y}_v], \tag{3}$$

the neighborhood term is a label-space KL divergence between a node's soft prediction and its neighborhood mean, $D_{\mathrm{neigh}}(v) = \mathrm{KL}\big(\hat{y}_v \,\|\, |N(v)|^{-1} \sum_{u\in N(v)} \hat{y}_u\big)$, and the view term measures active-view mode disagreement, $D_{\mathrm{view}}(v) = 1 - \max_c \frac{1}{M}\sum_m \mathbf{1}[\tilde{y}_v^m = c]$. D_{chain} is reserved for verified provenance/temporal chains and is not a driver of the reported results. No term uses clean labels or the noise mask; this is enforced by a regression test.

4.4 Evidence-preserving soft weighting

The evidence score protects samples that would be costly to discard, $e(v) = \mathrm{freq\_protect}(\tilde{y}_v)\,(1 + \gamma\,\mathrm{anom}(v))$, min-max normalized to $\tilde{e}(v)$. Rather than a small additive floor (which collapses to hard deletion), we use an evidence-*rescue* form so that suspicious yet high-evidence samples retain real weight:

$$w(v) = \underbrace{\sigma\big(-\kappa(\mathrm{GraphCDM}(v) - \theta)\big)}_{\text{clean branch}} + \underbrace{\big(1 - \sigma(\cdot)\big)\,\lambda_r\,\tilde{e}(v)}_{\text{evidence rescue}}. \tag{4}$$

The classifier minimizes $\sum_v w(v)\,\mathrm{CE}(f(z_v), \tilde{y}_v)$. Setting $\lambda_r = 0$ and thresholding recovers hard deletion; this is our `ablation_hard` comparator, sharing the identical graph, representation, and diagnostic so that the ablation isolates *only* the weighting.

We stress the intended interpretation of the evidence score: frequency protection encodes *deletion cost*, not trustworthiness. A rare sample is protected because discarding it is costly—it may be the only training evidence of a tail attack class—not because rarity implies a clean label. The score is thus a heuristic proxy for a risk-aware retention signal $R_{\mathrm{delete}}(v) \propto \mathrm{Cost}(\tilde{y}_v)\left(1 + \gamma\, \mathrm{anom}(v)\right)$, where Cost is high for rare attack classes, and the weighting in Eq. (4) implements risk-aware sample retention: samples with high diagnostic suspicion but high deletion cost keep reduced—never zero—influence. A full decision-theoretic treatment that models $P(\mathrm{clean} \mid x) \cdot \mathrm{Cost}(y)$ explicitly is left to future work; the empirical ordering in Section 6.3, where confident-learning-precision pruning outperforms our heuristic, indicates the value of sharpening the $P(\mathrm{clean} \mid x)$ component. Both the aligned CoLD baseline and the `ablation_hard` variant use the same shared encoder architecture, training budget, and view-masking configuration as Graph-CoLD; they differ only in the presence or absence of graph views and the weighting rule applied after Graph-CDM scoring.

4.5 Operational priority proxy

A priority score $P(v) = \alpha_1 \hat{y}_{\mathrm{mal}}(v) + \alpha_2 \mathrm{GraphCDM}(v) + \alpha_3 \tilde{e}(v)$ supports triage. We evaluate ranking indirectly and, as we show, Graph-CoLD ties strong peers on top-K precision; we therefore do not claim a ranking advantage.

4.6 Graph-CoLD Optimization Procedure

The complete Graph-CoLD pipeline consists of representation learning, label-space consistency estimation, evidence scoring, and weighted classifier optimization. Algorithm 1 summarizes the complete procedure.

Algorithm 1: Graph-CoLD evidence-preserving noisy-label learning.

Input: Multi-view graphs $\{G_m\}_{m=1}^{M}$, noisy labels $\tilde{Y}$, feature matrix X.

Output: Trained classifier f.

1. Construct active graph views according to available telemetry fields.
2. Learn view-specific representations using contrastive, temporal-alignment, and reconstruction objectives, then fuse them into node embeddings Z.
3. Generate prediction-space and neighborhood consistency statistics and compute a Graph-CDM score for each training sample.
4. Estimate evidence scores from label frequency and anomaly factors, then compute the evidence-preserving weights $w(v)$.
5. Optimize the classifier using weighted cross entropy and return f.

The design follows three considerations. First, the diagnostic operates in label and prediction space rather than pure embedding distance. This maintains consistency with the classifier decision process and avoids relying solely on representation similarity. Second, the weighting mechanism adopts an evidence-rescue formulation because suspicious samples may still contain useful attack information. A simple lower bound would approximate hard deletion, whereas the rescue formulation allows uncertain but informative samples to participate in training. Third, hard deletion is retained as the primary ablation because it isolates the contribution of evidence preservation while keeping the same graph representation and diagnostic components.

5 Experimental Design

5.1 Datasets and protocol

Table 1 summarizes the three real datasets. CICIDS-2017 is used under an 11-class *post-filter* policy with exact deduplication (24% of raw training rows removed; Section 6.4). CESNET-TLS-Year22 is a year-spanning backbone TLS dataset used under a 25-class audit-window policy (it replaces MALTLS-22, for which we lacked

Table 1 Datasets and active graph views. Source: `tables/table_1_dataset_protocol.csv`.

Dataset	Classes	Sample policy	Active views
CICIDS-2017	11	post-filter, exact-dedup	host/IP/temporal
CESNET-TLS-Year22	25	audit window	IP/temporal
UNSW-NB15	9	partition (no Worms)	temporal/behavioral

a verified license path). UNSW-NB15 is used under its partition with the smallest class (Worms) removed. To keep the study tractable and reproducible we evaluate deterministic audit windows (train 10,000 / test 5,000); we are explicit that this is a focused study, not a full-archive evaluation. Noise is injected only into training labels; we report symmetric, asymmetric, and graph-consistency noise, and for the rare-class analysis we use asymmetric noise concentrated on tail classes at rates $\{40, 60, 80\}\%$ with seeds 0–9.

All results in Tables 2–4 and Figures 2–3 are generated by the P2f clean de-oracled runner (`reports/p2f_tighten.md`) on these three datasets; earlier intermediate result files from D5/D9 used a different protocol and are not cited as final results.

Reproducibility and statistical protocol.

The audited data roots are local and are not committed to the repository. Each dataset uses deterministic audit windows of 10,000 training and 5,000 test instances. The tail analysis injects asymmetric noise at $\{40, 60, 80\}\%$ over seeds 0–9. Paired inference uses 30 like-for-like seed–rate observations per dataset (ten seeds by three noise rates), with Holm–Bonferroni correction and bootstrap 95% confidence intervals. The full command sequence is recorded in the reproduction commands of `reports/p2f_tighten.md`.

5.2 Baselines and metrics

We compare Graph-CoLD (soft and semi-supervised variants) with the aligned CoLD baseline and the `ablation_hard` deletion variant across the Tables 2–4 result matrix.

Two external noisy-label anchors were run under the *identical* P2f protocol (same audit windows, splits, seeds, and noise): Confident Learning (cleanlab `find_label_issues` with a shared-budget ExtraTrees head) and a full Co-Teaching implementation (dual MLPs with per-batch small-loss co-exchange and a forget-rate schedule tracking the true injected rate), recorded in `results/p5_external_baselines.csv` and compared in Section 6.3. Additional purification methods (MCRe, MORSE, FINE, Decoupling, and Co-Teaching-lite, a lightweight approximation superseded by the full implementation above) were evaluated under an earlier protocol. They were not re-run under the P2f clean-deduplication protocol and are therefore excluded from Tables 2–4 to avoid mixing incompatible result regimes; the MCRe/MORSE fidelity caveat is documented in `reports/p2b_baseline_fidelity.md`. Metrics reported here are aggregate Macro-F1, tail-class Macro-F1, tail recall, and paired statistical comparisons as specified above.

6 Results

6.1 RQ1: A label-space graph diagnostic modestly improves accuracy over CoLD

Table 2 reports per-dataset means. Graph-CoLD improves aggregate Macro-F1 over CoLD on all three datasets, with the gain concentrated on CICIDS, where host/IP/temporal views are most informative, and small on CESNET/UNSW. The improvements over CoLD are Holm-significant per dataset (CICIDS $p = 1.0 \times 10^{-3}$, CESNET $p = 4.1 \times 10^{-2}$, UNSW $p = 4.1 \times 10^{-2}$; source `tables/table_p2d_per_dataset_vs_cold.csv`). Notably, `ablation_hard` already captures most of this gain: the graph and the label-space diagnostic, not the weighting, drive aggregate accuracy.

Table 2 Per-dataset mean±std under high asymmetric tail noise (rates 40/60/80%, seeds 0–9; 30 scenarios per dataset). Sources: `tables/table_p2f_summary.csv` (Graph-CoLD family) and `results/p5_external_baselines.csv` (external anchors, same protocol). Best value per dataset and metric in bold.

Dataset	Method	Macro-F1	Tail-F1	Tail-recall
CICIDS-2017	CoLD	0.534±0.039	0.047±0.070	0.043±0.065
	ablation_hard	0.688±0.087	0.361±0.193	0.277±0.173
	Graph-CoLD-soft	0.721±0.086	0.434±0.191	0.328±0.180
	Graph-CoLD-semisup	0.687±0.093	0.357±0.206	0.272±0.181
	Confident-Learning	**0.769±0.048**	**0.605±0.103**	**0.517±0.112**
	Co-Teaching	0.567±0.071	0.118±0.152	0.128±0.171
CESNET-TLS-Year22	CoLD	0.658±0.130	0.411±0.247	0.306±0.211
	ablation_hard	0.680±0.132	0.453±0.249	0.331±0.218
	Graph-CoLD-soft	0.692±0.120	0.475±0.227	0.344±0.204
	Graph-CoLD-semisup	0.679±0.129	0.450±0.243	0.327±0.212
	Confident-Learning	**0.805±0.076**	**0.708±0.136**	**0.628±0.159**
	Co-Teaching	0.488±0.152	0.179±0.258	0.181±0.262
UNSW-NB15	CoLD	0.456±0.032	0.096±0.068	0.187±0.146
	ablation_hard	0.471±0.035	0.132±0.077	0.205±0.128
	Graph-CoLD-soft	0.485±0.033	0.173±0.078	0.229±0.120
	Graph-CoLD-semisup	0.469±0.033	0.130±0.074	0.194±0.119
	Confident-Learning	**0.499±0.026**	**0.254±0.085**	**0.317±0.123**
	Co-Teaching	0.419±0.036	0.055±0.080	0.120±0.178

6.2 Performance under Different Noise Types

To better understand when Graph-CoLD provides benefits, we analyze performance across different label-noise mechanisms. Table 3 summarizes Macro-F1 results under clean, symmetric, asymmetric, and graph-consistency noise.

The results show that the benefit of Graph-CoLD is strongly dataset-dependent. On CICIDS-2017, where host, IP, and temporal relations provide richer structural signals, Graph-CoLD achieves consistent improvements over CoLD across all noise types. Under asymmetric noise, the improvement is 0.107 in Macro-F1. This observation is consistent with the hypothesis that graph context is more valuable when corruption is associated with non-uniform class behavior.

Table 3 Graph-CoLD vs. CoLD Macro-F1 by noise type (mean over noise rates, seeds 0–9). Source: `tables/table_p4_noise_type_breakdown.csv`.

Dataset	Noise type	CoLD	Graph-CoLD-soft	Δ
CICIDS-2017	clean	0.671150	0.778689	0.107539
CICIDS-2017	symmetric	0.553919	0.592429	0.038510
CICIDS-2017	asymmetric	0.322779	0.429941	0.107162
CICIDS-2017	`graph_consistency`	0.496244	0.531555	0.035311
CESNET-TLS-Year22	clean	0.966321	0.968261	0.001941
CESNET-TLS-Year22	symmetric	0.898812	0.902585	0.003774
CESNET-TLS-Year22	asymmetric	0.698745	0.708302	0.009557
CESNET-TLS-Year22	`graph_consistency`	0.846591	0.849355	0.002764
UNSW-NB15	clean	0.494282	0.501909	0.007627
UNSW-NB15	symmetric	0.445324	0.445923	0.000599
UNSW-NB15	asymmetric	0.296512	0.303933	0.007421
UNSW-NB15	`graph_consistency`	0.434276	0.436233	0.001957

For CESNET-TLS-Year22 [38], the improvement is smaller because the baseline already achieves high performance under the evaluated protocol. Under this near-ceiling condition, additional graph information provides only marginal gains. UNSW-NB15 [39] shows a similar trend, indicating that graph denoising effectiveness depends on whether available views contain sufficient discriminative structure.

Overall, the noise-type analysis supports a bounded conclusion: Graph-CoLD is not universally superior under all conditions, but provides the largest benefit when the available relational views contain informative structure and when label corruption disrupts class-consistent evidence.

6.3 RQ2: Evidence preservation helps rare classes, not aggregate accuracy

The central finding concerns *where* evidence-preserving weighting matters. On aggregate Macro-F1, soft weighting is statistically indistinguishable from hard deletion (per-scenario Δ near zero, not significant). But on rare/tail attack classes, soft weighting significantly outperforms hard deletion (Table 4): ΔTail-F1 $= +0.073, +0.022, +0.040$ on CICIDS/CESNET/UNSW, all Holm-significant at $p < 10^{-4}$ with large effect sizes

Table 4 Powered tail-class comparison: Graph-CoLD-soft vs. hard deletion under asymmetric tail noise ($n = 30$ scenarios/dataset). Source: `tables/table_p2f_powered_tests.csv`.

Dataset	ΔTail-F1	Holm p	Cohen's d_z	Aggregate reg.?
CICIDS-2017	+0.0731	1.5×10^{-7}	1.28	none
CESNET-TLS-Year22	+0.0218	3.7×10^{-5}	0.91	none
UNSW-NB15	+0.0405	9.6×10^{-8}	1.29	none

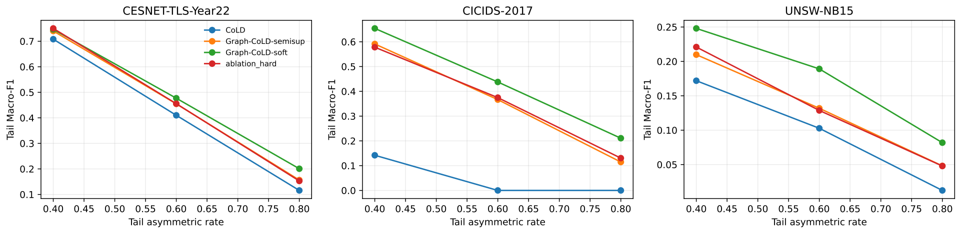

Fig. 2 Tail-class Macro-F1 of evidence-preserving soft weighting vs. hard deletion across asymmetric noise rates. The gain is small but consistent and significant on all three datasets.

($d_z = 1.28, 0.91, 1.29$), and with no significant aggregate-Macro-F1 regression. Figure 2 shows the effect is stable across noise rates. This is the operational point: hard deletion removes clean rare-attack evidence flagged as suspicious by the diagnostic; soft weighting keeps it at reduced weight, and the classifier recovers those classes.

We stress an honesty point. An earlier "rare-evidence recovery rate" that counted a sample as recovered whenever it was *retained* was tautological. We redefine recovery to require correct downstream classification; hard deletion is then no longer automatically zero. Because soft retention still scores near-ceiling on this diagnostic,[1] we rely solely on test-set tail-F1 for all headline comparisons.

External anchors under the identical protocol.

Table 5 places two external noisy-label methods, run under the identical P2f protocol (same audit windows, splits, seeds, and injected noise), against Graph-CoLD-soft with scenario-paired tests. The ordering is consistent across datasets. Confident Learning, a precise probabilistic pruner that removes only 7–24% of training rows, significantly

[1]The secondary diagnostic values and its definition are reported in `tables/table_p2f_summary.csv` and `reports/p2f_tighten.md`.

Table 5 External anchors vs. Graph-CoLD-soft under the identical P2f protocol: scenario-paired mean differences ($n = 30$ per dataset; positive = external method is better). All deltas shown are Holm-significant at $p < 0.05$ in the stated direction (Confident Learning above, Co-Teaching below Graph-CoLD-soft). Source: `tables/table_p5_external_tests.csv`.

Dataset	Method	ΔMacro-F1	ΔTail-F1	ΔTail-recall
CICIDS-2017	Confident-Learning	+0.048	+0.171	+0.189
	Co-Teaching	−0.155	−0.316	−0.201
CESNET-TLS-Year22	Confident-Learning	+0.114	+0.234	+0.283
	Co-Teaching	−0.204	−0.296	−0.163
UNSW-NB15	Confident-Learning	+0.014	+0.081	+0.089
	Co-Teaching	−0.066	−0.118	−0.109

outperforms Graph-CoLD-soft on aggregate and tail metrics on all three datasets. The evidence-preserving weighting in turn outperforms hard deletion (Table 4). Full Co-Teaching collapses on tail classes (mean Tail-F1 ≈ 0 at 60–80% noise on every dataset; `tables/table_p5_tail_by_rate.csv`), because its small-loss forgetting systematically discards the rare classes it should protect. Two lessons follow. First, the operational motivation of this study is confirmed empirically: deletion- and forgetting-based mechanisms can sacrifice rare-attack evidence (Co-Teaching) unless the diagnostic is precise (Confident Learning). Second, under the injected tail→benign noise model, precise pruning beats soft retention, because down-weighted flipped samples still teach the classifier that tail features indicate benign traffic. We therefore scope the evidence-preservation claim to the mechanism level—within the shared pipeline, soft weighting recovers rare classes that hard deletion loses—and we identify Confident Learning as the performance bar that graph-based denoising must clear; combining confident-learning-precision pruning with graph-structured diagnostics is future work. Per-noise-rate tail-F1 breakdowns for all methods are reported in `tables/table_p5_tail_by_rate.csv`.

6.4 RQ3: A leakage audit prevents inflated claims

Our initial pipeline reported a $\sim$ 28-point CICIDS advantage with a curve that barely degraded under 60% noise—implausible for a denoiser. Auditing revealed that CICIDS-2017 contains 152,201/630,399 ($\approx$ 24%) exact-duplicate training rows, and that any clean-label signal leaking into the evidence/graph construction inflates results. We verified zero train/test edge crossings, removed exact and near-duplicate edges, and eliminated the clean-label paths (enforced by a regression test). The apparent advantage collapses to +6.7 points on CICIDS, +0.3 on CESNET, and a slight -1.3 on UNSW in the audited comparison (`reports/p2c_leakage_audit.md`). These audit deltas quantify the magnitude of the inflation and were measured under the earlier P2c audit protocol (a 3,000/8,000 stratified audit sample for CICIDS and the D5.5 frozen matrix for CESNET/UNSW); they aggregate differently from, and are not directly comparable to, the final P2f result matrix in Tables 2–4 (`reports/p2f_tighten.md`). We report this because graph denoising in security is unusually susceptible to such inflation.

6.5 RQ4: Graph informativeness is consistent with when the method helps

A per-dataset neighborhood-homophily / view-coverage diagnostic is directionally consistent with Graph-CoLD's margin: CICIDS (rich host/IP/temporal views) shows the largest gain, CESNET sits near a performance ceiling, and UNSW (temporal/behavioral only) shows the smallest—occasionally negative—margin (Figure 3, tables/table_p2d_graph_informativeness.csv). With only three datasets this pattern is descriptive, not inferential. This turns the UNSW result into an informative boundary condition rather than a failure: when views lack structural signal, the graph diagnostic adds little.

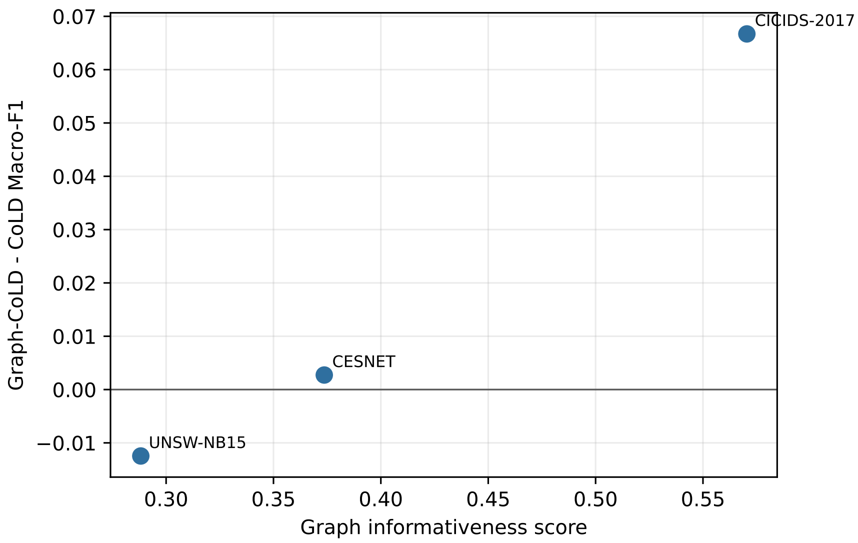

Fig. 3 Graph informativeness vs. Graph-CoLD margin over CoLD. Higher neighborhood homophily and view coverage are directionally consistent with larger gains (three datasets; descriptive pattern, not inferential).

6.6 RQ5: Cost

Graph-CoLD's overhead is moderate: mean runtime is comparable to CoLD and memory is lower than the strong purification baselines (the runtime and memory figures are measured under the earlier D5.5-protocol runs recorded in `results/runtime.json`; the relative pattern, not the absolute numbers, is the claim). Evidence preservation adds negligible cost over hard deletion because it changes only the training weights.

7 Discussion

From an operational perspective, Graph-CoLD is most suitable for SOC scenarios where multiple telemetry sources provide partially consistent evidence. Examples include environments where host relationships, communication patterns, and temporal behaviors jointly describe security events. In such settings, a suspicious sample does not necessarily indicate useless information. A rare attack instance may appear inconsistent because it represents an emerging technique rather than because it is incorrectly labeled.

However, the results also demonstrate clear boundaries. When available views are weak or incomplete, graph structure provides limited additional information. The UNSW-NB15 results illustrate this condition: without sufficiently informative relational views, graph-based diagnostics cannot reliably create additional evidence. Similarly, datasets operating near a performance ceiling, such as CESNET-TLS-Year22 under the evaluated protocol, naturally limit the observable improvement.

A key methodological decision in this work is the avoidance of unfair state-of-the-art comparisons. Two external anchors—Confident Learning and a full Co-Teaching implementation—were therefore run under the identical leakage-audited protocol and are reported in Table 5, even though one of them outperforms our method on every dataset; that result is part of the evidence, not an inconvenience. Other purification and robust-training variants (MCRe, MORSE, FINE, Decoupling) were not all reimplemented under the final leakage-audited protocol. Mixing results obtained under different data-cleaning, duplication-removal, and evaluation settings would create an unreliable comparison, so those methods remain excluded from the formal matrix. A comprehensive comparison with all previous methods requires faithful implementations or official code under a unified evaluation framework.

The external-anchor ordering carries a practical lesson for SOC deployments. Confident Learning's precise pruning—removing only 7–24% of rows—sets the performance bar under the injected noise model, while Co-Teaching's aggressive small-loss forgetting eliminates the rare attack classes almost entirely. The differentiator is not deletion versus retention but *diagnostic precision*: an imprecise purifier sacrifices exactly the evidence an analyst needs, whether it deletes it (Co-Teaching) or down-weights it insufficiently (Graph-CoLD under tail$\rightarrow$benign flips). Combining confident-learning-precision pruning with graph-structured diagnostics is the natural next step.

The leakage audit provides a broader methodological lesson for graph-based security machine learning. Graph models can unintentionally exploit duplicated records, cross-split connections, or hidden clean-label information. Such effects may produce impressive but misleading improvements. Therefore, graph security studies should explicitly verify data separation, graph construction constraints, and label independence before reporting performance gains.

Future work should extend Graph-CoLD toward streaming SOC environments, full-archive evaluation, analyst-in-the-loop validation, and broader cross-domain benchmark comparisons. These extensions are necessary to determine whether evidence-preserving denoising remains effective when labels arrive continuously and attack distributions evolve over time. The results support a bounded but useful claim. A label-space graph diagnostic modestly improves noisy-label IDS over CoLD where views are informative, and—more distinctively—evidence-preserving weighting protects rare-attack evidence that hard deletion discards, improving tail-class detection under high asymmetric noise at no aggregate cost *within the shared pipeline*; under the same protocol, confident-learning pruning remains the stronger performer and marks the bar for future graph-based denoisers. In SOC terms, this matters precisely for the samples an analyst cannot afford to lose: rare or emerging attacks whose labels are most likely to be noisy. We deliberately avoid two overclaims that our own experiments do not support: a universal accuracy advantage (the aggregate gain is modest and largely from the graph, not the weighting) and a ranking advantage (Graph-CoLD ties strong peers on top-K precision).

8 Threats to Validity

Internal. The evaluation uses deterministic audit windows for tractability; absolute Macro-F1 is therefore lower than full-archive runs, though the paired, seed-controlled comparisons are the object of inference. MCRe and MORSE pass a clean-label sanity floor but degrade on high-noise tabular settings more than in their original domains; we report this caveat rather than a possibly unfaithful number (`reports/p2b_baseline_fidelity.md`). *Construct.* The rare-evidence-recovery diagnostic, even after correction, saturates for soft retention; we therefore headline test-set tail-F1. *External.* CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15 are useful benchmarks but not complete SOC deployments; the priority score is evaluated by proxy, not an analyst study. The two-dataset lock recorded in the D9 submission audit was an intermediate checkpoint; the authoritative final scope is the three-dataset P2f clean de-oracled protocol.

9 Limitations

MALTLS-22 and OpTC are not evaluated (no verified license/provenance path). Behavioral and threat-intelligence views are disabled where fields are absent. The aggregate accuracy contribution is modest and dataset-dependent; under the audited protocol, Confident Learning outperforms Graph-CoLD on aggregate and tail metrics on all three datasets, so the distinctive contribution is the leakage audit, the mechanism-level evidence-preservation finding, and the informativeness boundary analysis rather than state-of-the-art accuracy. Full-archive and streaming evaluations are left to future work.

10 Conclusion

We revisited collaborative label denoising for intrusion detection and asked, with unusual auditing discipline, whether a multi-view label-space graph diagnostic and evidence-preserving weighting improve robustness. The honest answer is targeted: the graph diagnostic gives a modest, informativeness-dependent accuracy gain over CoLD, and evidence-preserving weighting significantly improves rare-attack tail-F1 over hard deletion under high asymmetric noise at no aggregate cost across three real datasets—while a confident-learning anchor, evaluated under the identical protocol, outperforms both, and full co-teaching sacrifices the rare classes outright. Alongside, we contribute a leakage audit and a graph-informativeness diagnostic that together explain when—and when not—graph denoising helps. More broadly, our results argue that security data curation should shift from indiscriminate purification toward evidence-aware retention, in which deletion decisions explicitly weigh the operational cost of discarding rare-attack evidence against the precision of the noise diagnostic. We hope the audit, and the audited external-baseline ordering, are as useful to the community as the method.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used generative AI assistants (Kimi, Moonshot AI; GPT-based coding assistants) in order to assist with code development and debugging of the experimental pipeline, and to assist with language editing and drafting of the manuscript and submission documents. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. The tools were not used to generate, alter, or analyze research data, and all experimental results and statistical claims were verified by the authors against the frozen experimental artifacts.

Availability of data and materials

All three datasets are publicly available from their original providers: CIC-IDS-2017 from the Canadian Institute for Cybersecurity, University of New Brunswick; CESNET-TLS-Year22 from Zenodo (DOI: 10.5281/zenodo.10608607); and UNSW-NB15 from the University of New South Wales, Canberra. The derived artifacts required to reproduce our results—code, configuration files, frozen data-split manifests with content hashes, result tables, and figure-generation scripts—will be released in a public repository upon acceptance of this manuscript.

Authors' contributions

Provided in the separate title page, in accordance with the journal's double-anonymous review policy.

Funding

No specific funding was received for this work.

Competing interests

The authors declare that they have no competing interests.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

References

[1] S. Yang, X. Zheng, J. Li, J. Xu, E. C. H. Ngai, CoLD: Collaborative label denoising framework for network intrusion detection, in: Network and Distributed System Security (NDSS) Symposium, 2026, to appear. Available: https://dx.doi.org/10.14722/ndss.2026.231950

[2] D. Arp et al., Dos and don'ts of machine learning in computer security, in: USENIX Security Symposium, 2022, pp. 3971–3988.

[3] H. Song, M. Kim, D. Park, Y. Shin, J.-G. Lee, Learning from noisy labels with deep neural networks: A survey, IEEE TNNLS 34 (11) (2022) 8135–8153.

[4] G. Patrini, A. Rozza, A. K. Menon, R. Nock, L. Qu, Making deep neural networks robust to label noise: A loss correction approach, in: CVPR, 2017, pp. 1944–1952.

[5] B. Han et al., Co-teaching: Robust training of deep neural networks with extremely noisy labels, in: NeurIPS, 2018, pp. 8535–8545.

[6] X. Yu et al., How does disagreement help generalization against label corruption?, in: ICML, 2019, pp. 7164–7173.

[7] E. Malach, S. Shalev-Shwartz, Decoupling "when to update" from "how to update", in: NeurIPS, 2017, pp. 960–970.

[8] J. Li, R. Socher, S. C. H. Hoi, DivideMix: Learning with noisy labels as semi-supervised learning, in: ICLR, 2020.

[9] C. G. Northcutt, L. Jiang, I. L. Chuang, Confident learning: Estimating uncertainty in dataset labels, JAIR 70 (2021) 1373–1411.

[10] T. Kim, J. Ko, J. Choi, S.-Y. Yun, FINE samples for learning with noisy labels, in: NeurIPS, 2021, pp. 24137–24149.

[11] X. Wu, W. Guo, J. Yan, B. Coskun, X. Xing, From grim reality to practical solution: Malware classification in real-world noise, in: IEEE S&P, 2023, pp. 2602–2619.

[12] Q. Yuan, G. Gou, Y. Zhu, Y. Zhu, G. Xiong, Y. Wang, MCRe: A unified framework for handling malicious traffic with noise labels, IEEE Transactions on Information Forensics and Security, 2023.

[13] T. N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, in: ICLR, 2017.

[14] P. Veličković et al., Graph attention networks, in: ICLR, 2018.

[15] W. L. Hamilton, R. Ying, J. Leskovec, Inductive representation learning on large graphs, in: NeurIPS, 2017.

[16] Z. Hu, Y. Dong, K. Wang, Y. Sun, Heterogeneous graph transformer, in: The Web Conference (WWW), 2020, pp. 2704–2710.

[17] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, A simple framework for contrastive learning of visual representations, in: ICML, 2020, pp. 1597–1607.

[18] M. U. Rehman, H. Ahmadi, W. U. Hassan, Flash: A comprehensive approach to intrusion detection via provenance graph representation learning, in: IEEE S&P, 2024, pp. 3552–3570.

[19] J. Xu, X. Shu, Z. Li, Understanding and bridging the gap between unsupervised network representation learning and security analytics, in: IEEE S&P, 2024, pp. 3590–3608.

[20] L. Yang et al., CADE: Detecting and explaining concept drift samples for security applications, in: USENIX Security Symposium, 2021, pp. 2327–2344.

[21] R. Vaarandi, A. Guerra-Manzanares, Network IDS alert classification with active learning techniques, Journal of Information Security and Applications 81 (2024) 103687.

[22] L. Chavali, A. Krishnan, P. Saxena, B. Mitra, A. Sreevallabh Chivukula, Off-policy actor-critic deep reinforcement learning for alert prioritization in intrusion detection systems, Computers & Security 142 (2024) 103854.

[23] I. Sharafaldin, A. H. Lashkari, A. A. Ghorbani, Toward generating a new intrusion detection dataset and intrusion traffic characterization, in: ICISSP, 2018, pp. 108–116.

[24] K. Hynek, J. Luxemburk, J. Pesek, T. Cejka, P. Siska, CESNET-TLS-Year22: A year-spanning TLS network traffic dataset from backbone lines, Scientific Data 11 (2024) 1156.

[25] N. Moustafa, J. Slay, UNSW-NB15: A comprehensive data set for network intrusion detection systems, in: Military Communications and Information Systems Conference (MilCIS), 2015, pp. 1–6.

[26] J. L. Guerra, C. Catania, E. Veas, Datasets are not enough: Challenges in labeling network traffic, Computers & Security 120 (2022) 102810.

[27] Y. Qing et al., Low-quality training data only? A robust framework for detecting encrypted malicious network traffic, arXiv preprint arXiv:2309.04798, 2023.

[28] Z. Zhao et al., Enhancing robustness of on-line learning models on highly noisy data, IEEE TDSC 18 (5) (2021) 2177–2192.

[29] N. Wang et al., FeCo: Boosting intrusion detection capability in IoT networks via contrastive learning, in: IEEE INFOCOM, 2022, pp. 1409–1418.

[30] Y. Yue et al., Contrastive learning enhanced intrusion detection, IEEE TNSM 19 (4) (2022) 4232–4247.

[31] Y. Liu et al., DeepDRAC: Disposition recommendation for alert clusters based on security event patterns, IEEE TIFS 20 (2025) 6443–6458.

[32] H. G. Shon, Y. Lee, M. Yoon, Semi-supervised alert filtering for network security, Electronics 12 (23) (2023) 4755.

[33] W. Wang et al., Transformer-based framework for alert aggregation and attack prediction in a multi-stage attack, Computers & Security 136 (2024) 103533.

[34] X. Xia et al., Sample selection with uncertainty of losses for learning with noisy labels, arXiv preprint arXiv:2106.00445, 2021.

[35] C. Liu et al., Noise-resistant deep metric learning with ranking-based instance selection, in: CVPR, 2021, pp. 6811–6820.

[36] S. Yang et al., RecDA: Concept drift adaptation with representation enhancement for network intrusion detection, in: ACM SIGKDD, 2024, pp. 3818–3828.

[37] G. Andresini et al., Insomnia: Towards concept-drift robustness in network intrusion detection, in: ACM AISec, 2021, pp. 111–122.

[38] K. Hynek et al., CESNET-TLS-Year22 dataset, Zenodo record 10608607, 2024.

[39] N. Moustafa, J. Slay, The evaluation of network anomaly detection systems: UNSW-NB15, Information Security Journal 25 (1–3) (2016) 18–31.